

SHINING : Conception and Implementation of a Do-It-Yourself Electroplating Machine Project intended for individual use (makers).

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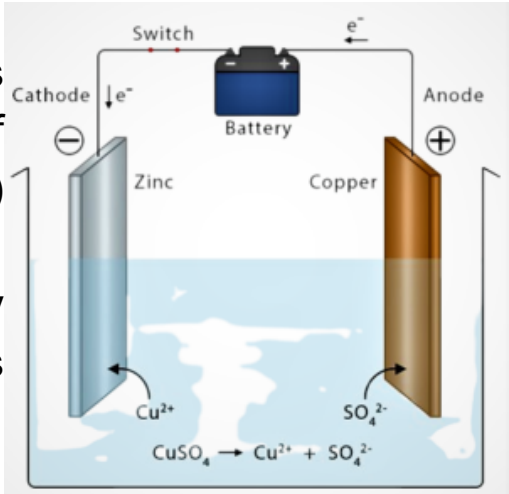
Description : Shining is an all in one DIY electroplating machine that answers those constraints :

- It can be used without an external generator
- With consistent results even with minimum knowledge of the practice
- As safe as possible to use
- Is Fully made with only generic parts and a standard 3D printer

What's electroplating ?

Electroplating or electroforming consists of using electricity to deposit a thin layer of metal (copper, nickel, palladium, gold...) on a conductive piece.

Electroplating can be used in many diy fields like jewelry, electronics or miniatures painting



Why electroplating for makers/diy ?

Electroplating can be used by makers to produce parts with metallic characteristics such as :

- electric conductivity
- mechanical properties (strength, durability)
- look and feel

This process also relevant for its accessibility by being way less complex to implement than traditional machining/molding

Also, since only the top layer is covered, it only uses a fraction of the metal which makes the parts cheaper to make

What am I proposing ?

This system (shining) aims at being as simple, cheap, and robust as possible.

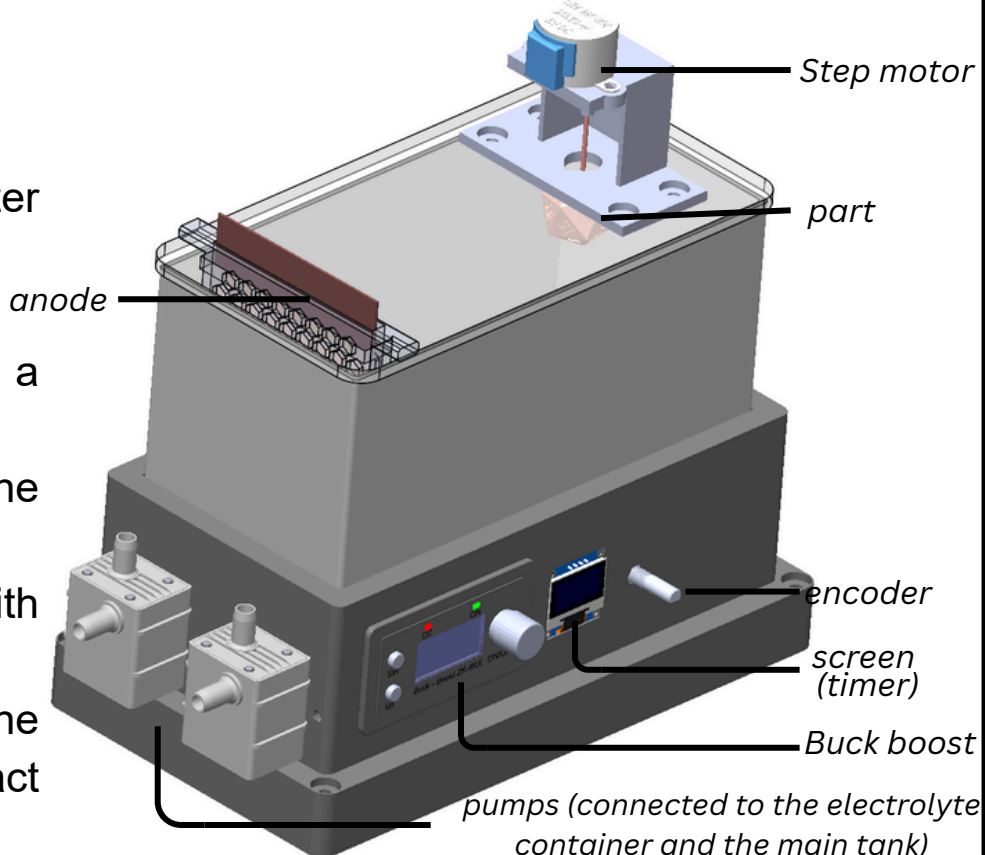
It can be totally assembled for less than 150€ (electrolyte solution aside) with a standard 3D printer and a few generic part (arduino uno, 5V step motor, 12V pumps, encoder, oled screen...)

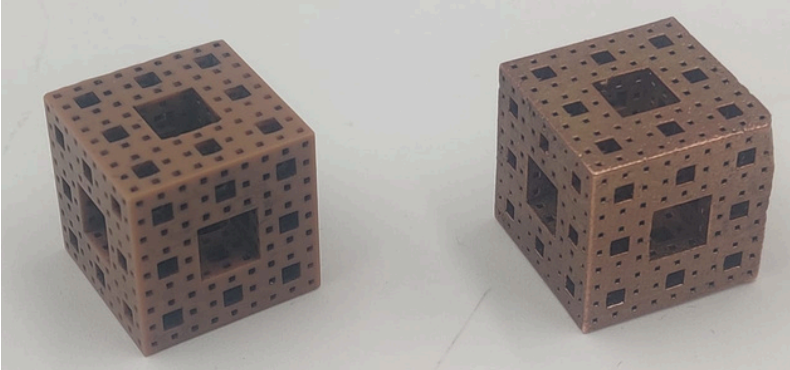
It works around a Buck Boost Zk-4KX (regularly used to make diy power supplies) which provide a constant current between the part and the anode (copper, nickel, zinc...)

To use it, the user attach the part (connected to the - of the supply) to the step motor and connect the anode (+ of the supply).

They then enter the plating duration with a simple encoder which activated a pump to fill the tank with the acidic electrolyte (closing the circuit and beginning the plating).

The part is rotated in order to assure uniform plating. When the timer ends (the plating stops), the electrolyte solution is sucked out by a second pump and replaced in its container (reducing contact with the user)





before an after plating (without sanding)



before plating after plating and sanding

My results (so far) :

What works :

- + the parts are evenly plated with a thick coating (+30% in weight)
- + strong bond between the part and the metal (no detachment observed)
- + the plating makes the parts utterly conductive in a way that isn't attainable with conductive paint

Whats left to improve

- the parts come out rough and need some sanding for a smooth finish
- attaching the part to the step motor and can be a little fiddly
- only works for small parts (> 9 cm in diameter)
- test needs to be done to evaluate the thickness of the coating in order to adjust tolerances for mechanical parts
- a fume extractor is needed to make the system safe in places with not enough ventilation